Scenario 3: R Markdown to PDF

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2026-08-31

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1 What this file is

This is the workflow DSCI 551 labs use: write prose and R code in one `.Rmd` file, then render straight to PDF.

Three things to notice: chunk options control what reaches the PDF, inline code lets you put computed numbers inside sentences, and the whole document runs top-to-bottom in a **fresh** R session when knitted.

2 Chunk options

The default: code shown, output shown.

```r
summary(mtcars$mpg)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   10.40   15.43   19.20   20.09   22.80   33.90
```

`echo=FALSE` hides the code but keeps the output. Lab questions that ask for a plot usually want this.

```
##   cylinders cars
## 1         4   11
## 2         6    7
## 3         8   14
```

`eval=FALSE` shows the code but does not run it.

```r
# This line is never executed.
install.packages("something")
```

3 Inline code

The `mtcars` dataset has 32 rows and 11 columns. Mean fuel economy is 20.09 mpg.

4 A figure

```
ggplot(mtcars, aes(x = wt, y = mpg, colour = factor(cyl))) +
  geom_point(size = 2.5) +
  labs(x = "Weight (1000 lbs)", y = "Miles per gallon", colour = "Cylinders") +
  theme_minimal()
```

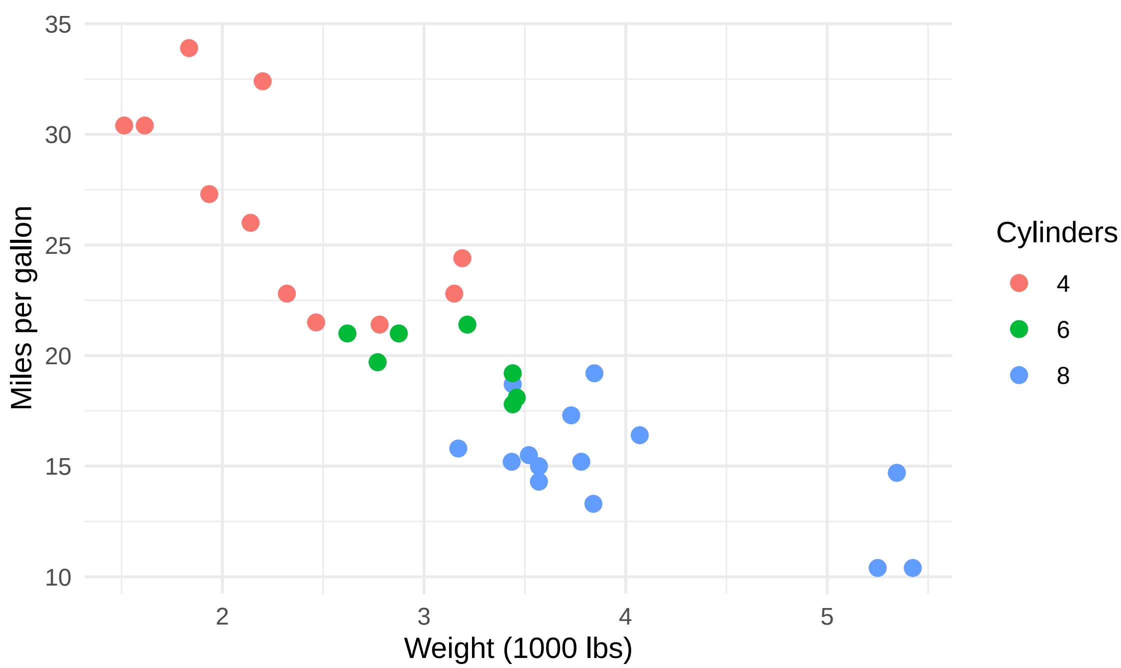

Figure 1: Weight versus fuel economy, coloured by cylinder count.

5 A table

```
knitr::kable(
  head(mtcars[order(-mtcars$mpg), c("mpg", "cyl", "wt", "hp")], 5),
  caption = "Five most fuel-efficient cars."
)
```

Table 1: Five most fuel-efficient cars.

	mpg	cyl	wt	hp
Toyota Corolla	33.9	4	1.835	65
Fiat 128	32.4	4	2.200	66
Honda Civic	30.4	4	1.615	52
Lotus Europa	30.4	4	1.513	113
Fiat X1-9	27.3	4	1.935	66

6 Math

LaTeX math works because the PDF goes through LaTeX. DSCI 551 will need this constantly.

Inline: the mean is $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$.

Display:

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta} \prod_{i=1}^{n} f(x_i \mid \theta)$$

7 The one trap

Knitting runs the document in a **brand new R session**. Anything you created interactively in your terminal does not exist there.

So a document can work perfectly while you click through it, and still fail to knit. Before submitting, always knit once from scratch and check the PDF actually came out.

```
# This works when knitted because tidyverse is loaded in the setup chunk
# above. If it were only loaded in your terminal, knitting would fail.
mtcars |> summarise(n = n(), mean_mpg = mean(mpg))
```

```
##    n mean_mpg
## 1 32 20.09062
```